

ACS test client user guide

The `test-client` is a command line application that can be used to test the ACS and the connection to the DAP. This application can be very useful during the integration of a new project or when custom scripting is needed to fulfil a task.

A description of the available commands can be found by starting the test-client with the flag `-h` (for help).

```
$ ./test-client.exe -h
ACS test client
Usage: test-client [OPTIONS] SUBCOMMAND

Options:
  -h, --help                Print this help message and exit
  -p, --port INT REQUIRED    Port of ACS server
  -a, --address TEXT        IP address of ACS server

Subcommands:
  get_capabilities
  get_dap_version
  get_device_info
  get_dsp_restrictions
  get_log_sweep_info
  get_parameter_ids
  get_parameter_info
  get_parameter_values
  get_parameter_values_normalized
  get_pink_noise_info
  get_rtcore_version
  get_stimuli_gain_info
  get_stimuli_info
  prepare_stimuli
  set_active_configuration_by_name
  set_active_mode
  set_archive
  set_bypass_gain
  set_configuration
  set_module_parameters
  set_parameter_values
  set_parameter_values_normalized
  set_processing_enabled
  set_stimuli_gain
  start_stimuli
  test_notifications
  test_parameter_notification
```

Additional help is available for each subcommand.

Examples

Note that the port number used in these examples is 55813 which is taken from the example-config.json file.

Get device info

```
./test-client.exe -p 55813 get_device_info
Got DeviceInfo:
  Manufacturer:      Dirac Research
  Name:              Test
  Model name:        Tuning ACS
  Unique device id:
  Number of speakers: 0
  Number of microphones: 0
  Number of filter slots: 0
  System name:
```

Set a module parameter

```
./test-client.exe -p 55813 set_module_parameters 1 2 2.12
Sending ModuleParamValues:
(0x1, 2) -> 2.12
```

Update the gain of EQ band 0 to -6.0dB

```
./test-client.exe -p 55813 set_parameter_values 0x100 -6.0
```

Load a DAR file

```
./test-client.exe -p 55813 set_archive tuning.dar
```

Instantiate a tuning configuration in the active (loaded) DAR file

```
./test-client.exe -p 55813 set_active_configuration_by_name "MyConfiguration"
```

Mimicking Filterlab measurements

This section demonstrates how to use the ACS Client to mimic the behavior of Dirac FilterLab to carry out a measurement.

Note that there might be differences in sequence and timing when a real measurement is done from Filterlab

- Load the DAR file that includes the measurement configuration

```
./test-client.exe -p 55813 set_archive poc_05_measurement_setup_8x8.dar
./test-client.exe -p 55813 set_active_configuration_by_name ""
```

- Mimic the calls traditionally performed by Dirac FilterLab to prepare for a measurement

```
./test-client.exe -p 55813 get_device_info
./test-client.exe -p 55813 get_stimuli_info
./test-client.exe -p 55813 get_log_sweep_info 0
./test-client.exe -p 55813 get_pink_noise_info 0
```

- Mimic the calls traditionally performed by Dirac FilterLab to find a suitable gain level for the actual measurement. Pink Noise will be used for this exercise. Below commands can be repeated for each channel.

Due to the simple design of the ACS client, it is not possible to send commands after the stimuli generation has been started. Therefore, the order of the commands below differs compared to how this is used by FilterLab.

```
# Tell the Dirac runtime module to output Pink Noise (arg1=1) on channel 4
# (arg2=3)
./test-client.exe -p 55813 prepare_stimuli 1 3

# Adjust the gain of the active channel (set gain of channel 4 to -40dB)
./test-client.exe -p 55813 set_stimuli_gain 3 -40

# Activate the Pink Noise
./test-client.exe -p 55813 start_stimuli

# Cancel the sound emission
# Due to the simple design of the ACS client, this is only possible by
# aborting the process
Ctrl + C
```

Note on set_stimuli_gain: the input gain value is expressed in decibels. The allowed range is <-40dB, 0dB>.

- Mimic the calls traditionally performed by Dirac FilterLab to perform an actual measurement using a logarithmic sweep

```
# Tell the Dirac runtime module to output a logarithmic sweep (arg1=3)
# beginning the output sequence from the first channel (arg2=0)
./test-client.exe -p 55813 prepare_stimuli 3 0

# Adjust the gain of each channel (the gain is expressed in decibels and
# it must be the same for all channels)
```

```
./test-client.exe -p 55813 set_stimuli_gain 0 -20  
./test-client.exe -p 55813 set_stimuli_gain 1 -20  
./test-client.exe -p 55813 set_stimuli_gain 2 -20  
./test-client.exe -p 55813 set_stimuli_gain 3 -20  
./test-client.exe -p 55813 set_stimuli_gain 4 -20  
./test-client.exe -p 55813 set_stimuli_gain 5 -20  
./test-client.exe -p 55813 set_stimuli_gain 6 -20  
./test-client.exe -p 55813 set_stimuli_gain 7 -20
```

```
# Begin the measurement
```

```
./test-client.exe -p 55813 start_stimuli
```

```
# The measurement will complete automatically and release the process
```